

Frieze and Wallpaper Symmetry Exploration in Cultural Artifacts

Objective: Explore the mathematical beauty of symmetry in cultural artifacts by identifying and analyzing frieze and wallpaper patterns. This assignment will help you apply concepts of symmetry groups to real-world examples from non-Western cultural contexts.

Available Resources: In addition to web resources, you are expected to use Chapter 28 of our course textbook and the article “Symmetries of Culture” available on the course Moodle page.

Part 1—Presentation (In-Class): Choose a non-Western culture and find examples of 3 different frieze symmetries and 2 different wallpaper symmetries. For each of these, you will identify and label the correct symmetry group according to standard frieze and wallpaper classification.

For your presentation, use slides or physical examples to show your patterns. Explain

1. The cultural context and origin of each artifact.
2. The specific symmetry type of each pattern.
3. Why each pattern belongs to the symmetry group you have identified.

In addition, each group will present a problem from the Chapter 28 exercises. These will be assigned at a later date.

Your presentation should be between 10 and 12 minutes. Be sure to cite your sources in your presentation.

Part 2—Paper Submission: Each person (not each group) will write a short paper (2 - 3 pages) that serves as a brief reflection on your chosen patterns and your presentation. How did this assignment help you see the connections between symmetry in mathematics and cultural art? Two paragraphs will reflect on your presentation and partner work.

Evaluation Criteria: A rubric will be posted in the Course Project area on Moodle. Your presentation will be evaluated on accuracy in identifying symmetry groups, depth of cultural understanding, and clarity in explanation. Your paper will be evaluated on the quality of reflection and your demonstration of your understanding of symmetry concepts.